

Label Set

Unit of analysis (text unit) = entire review

Dimension	Classes/Labels	Granularity
Emotions (13)	Joy, Love, Surprise, Contentment, Interest, Anger, Frustration, Disappointment, Sadness, Fear, Disgust, Boredom, None	Multi-label per text unit
Commitment	Low / Medium / High	Ordinal per text unit
Recommendation	Would not / Neutral / Would recommend	Ordinal per text unit

Definitions and Decision Rules

1. Emotions: What the user feels about the book. Each review must be assigned at least one of the 13 following labels.
 - a. Joy: the user expresses happiness/enjoyment/amusement/mirth (humor)
 - b. Love: the user expresses love or uses very complimentary language
 - c. Surprise: the user is shocked by something, or finds something unexpected
 - d. Contentment: the user expresses contentment/satisfaction; says they “like” something about the book
 - e. Interest: the user is intrigued/captivated/pensive
 - f. Anger: the user is mad
 - g. Frustration: the user expresses frustration/annoyance/irritation, but not quite anger
 - h. Disappointment: the user expresses disappointment
 - i. Sadness: the user expresses sadness or some other negative feeling that doesn’t fit into any other categories (such as guilt); says the book or some aspect of it made them “emotional”
 - j. Fear: the user expresses fear or a feeling of being creeped out/unsettled
 - k. Disgust: the user expression repulsion/distaste
 - l. Boredom: the user expresses boredom/disinterest
 - m. None: no emotions are detected (default category)
2. Commitment: how likely the user is to continue reading other books in the series and/or other books by the author. Each review must be given one and only one of the following labels.
 - a. Low: the user is *unlikely* to continue reading other books in the series and/or other books by the author
 - b. Medium: *it is unclear* whether the user will continue reading other books in the series and/or other books by the author (default category)

- c. High: the user is *likely* to continue reading other books in the series and/or other books by the author
- 3. Recommendation: how likely the user is to recommend the book. Each review must be given one and only one of the following labels.
 - a. Would not: the user is *unlikely* to recommend the book
 - b. Neutral: *it is unclear* whether the user would recommend the book (default category)
 - c. Would recommend: the user is *likely* to recommend the book

Borderline Cases

- 1. If a review is unintelligible assign the following default labels: none (emotion), medium (commitment), neutral (recommendation)

“None/Other” Policy

- 1. Emotion: default to none
- 2. Commitment: default to medium
- 3. Recommendation: default to neutral

Example Texts

- 1. Emotions: What the user feels about the book. Each review must be assigned at least one of the 12 following labels.
 - a. Joy:
 - i. This book made me so happy!
 - ii. It put a smile on my face
 - b. Love:
 - i. I love the main character
 - ii. I adore the writing style
 - c. Surprise:
 - i. I was not expecting that ending
 - ii. So many twists and turns!
 - d. Contentment:
 - i. What a peaceful read
 - ii. I really enjoyed this book
 - e. Interest:
 - i. What an interesting read
 - ii. The third chapter was intriguing
 - f. Anger:
 - i. Reading this made my blood boil
 - ii. I hated the main character
 - g. Frustration:

- i. Reading this book made me so annoyed
 - ii. I am frustrated
 - h. Disappointment:
 - i. I was really hoping that the ending would be different
 - ii. What a disappointing read
 - i. Sadness:
 - i. I cried a lot
 - ii. What a disheartening book
 - j. Fear:
 - i. I was so creeped out the whole time
 - ii. How scary
 - k. Disgust:
 - i. The main character was repulsive
 - ii. The gore was disgusting
 - l. Boredom:
 - i. Meh, I didn't really feel anything
 - ii. I was uninterested and didn't finish reading
 - m. None:
 - i. sd;lkjf
 - ii. My
- 2. Commitment: how likely the user is to continue reading other books in the series and/or other books by the author. Each review must be given one and only one of the following labels.
 - a. Low:
 - i. I will never read another book by this author
 - ii. I can't bring myself to finish it
 - b. Medium:
 - i. I can't decide whether I will finish the series
 - ii. I may or may not read more books by this author
 - c. High:
 - i. I can't wait for the sequel's release!
 - ii. I will read every book that this author writes
- 3. Recommendation: how likely the user is to recommend the book. Each review must be given one and only one of the following labels.
 - a. Would not:
 - i. Do not read this
 - ii. Save yourself the time and read something else
 - b. Neutral:
 - i. If you've got extra time, maybe try this book
 - ii. I'm not sure whether this book is worth reading

- c. Would recommend:
 - i. You must read this book
 - ii. I highly recommend reading this

Data Source Justification

The source for this project is the UCSD Goodreads dataset, which contains large-scale user reviews pulled from Goodreads and made available through a public dataset page. This source contains the construct of interest because the review text includes the emotional language, recommendation signals, and continuation cues needed to classify emotions, recommendation intent, and commitment.

Source fit and business relevance

- Goodreads reviews are a strong proxy for the kind of customer feedback Underbrush would want to analyze because they directly reflect how readers feel about books, whether they would recommend them, and whether they want to continue with the author or series. This makes the dataset well suited for studying which emotional responses are associated with highly rated books and for improving personalized book recommendations.

Operational use in a firm

- Underbrush would use the Goodreads dataset as a recurring training and benchmarking resource to understand patterns in reader language, then periodically retrain and validate the model as new labeled data becomes available. In practice, the dataset would support model development, performance checks, and quarterly refreshes, while the deployed model could then be applied on an ongoing basis to newly scraped user reviews to monitor emotion, recommendation intent, and commitment trends over time.

Access plan

- We will access the dataset through the provided JSON download from the UCSD Goodreads page rather than scraping the site directly, following the site's access

rules. For ongoing use, new review data scraped from a relevant review page will be run through the same pipeline to continuously evaluate incoming reviews.

Sampling Plan

To ensure the dataset reflects the full range of the construct, we will use a stratified sampling approach when selecting reviews for annotation and model development. Reviews will be sampled across rating levels, review lengths, and book categories so that the training data includes both highly emotional and more neutral examples, as well as a mix of positive and negative sentiment. This will help capture rare emotions and improve the model's ability to generalize to new user reviews. A separate held-out test set will be reserved for final evaluation so that performance can be measured on unseen data.